

Instructions: There are totally 2 problem2. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Given the following autonomous differential equation:

$$\frac{dx}{dt} = x^2 - 4x.$$

Find all critical points of the given differential equation and then determine whether it is stable or unstable for each of them.

Solution $f(x, t) = x^2 - 4x = 0 \Rightarrow x(x-4) = 0$
 $x=0, x=4 \rightarrow$ two critical point

$$\begin{array}{ccccc} x' > 0 & & x' < 0 & & x' > 0 \\ \rightarrow & & \leftarrow & & \rightarrow \\ & x=0 & & x=4 & \end{array}$$

$$x=0 \text{ stable}$$

$$x=4 \text{ unstable}$$

Problem 2. (5 points) Suppose that a motorboat is moving at 60 ft/s when its motor suddenly quits at time $t = 0$, and that 10 s later the boat has slowed to 30 ft/s. Assume that the resistance it encounters while coasting is proportional to its velocity, i.e., $\frac{dv}{dt} = -kv$. Find the velocity $v(t)$ of the motorboat at time t .

Solution: $\frac{dv}{dt} = -kv \rightarrow$ natural growth model $\Rightarrow v(t) = Ae^{-kt}$

$$v(0) = 60 \Rightarrow Ae^0 = 60 \Rightarrow A = 60 \Rightarrow v(t) = 60e^{-kt}$$

$$v(10) = 30 \Rightarrow 60e^{-10k} = 30$$

$$e^{-10k} = \frac{1}{2} \Rightarrow -10k = \ln\left(\frac{1}{2}\right)$$

$$k = \frac{\ln\left(\frac{1}{2}\right)}{-10} = \frac{\ln 2}{10}$$

$$\text{thus } v(t) = 60e^{-\left(\frac{\ln 2}{10}\right)t}$$